

Course Title	Applied Calculus	Course Code	MT1001
Department	Department of Electrical Engineering (DEE)	Campus	Lahore
Knowledge Profile	Mathematics & Computing (WK2)	Credit Hrs.	3
Knowledge Area	Mathematics (KA10)	Grading Scheme	To be announced by the instructor
HEC Knowledge Area	Natural Sciences	Applicable From	Fall 2022
Pre-requisite(s)			

Course Objective	Develop a sound understanding of basic concepts of calculus and its applications
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No.	Assigned Program Learning Outcome (PLO)
2	An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering science.

I = Introduction, R = Reinforcement

E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Solve the algebraic equations and inequalities.	Q1, M1	C3	2.1H, 2.3H
2	Identify and analyze the function.	A1, M1	C4	2.2H, 2.4L
3	Compute the derivative of a function and apply the concept of differentiation in engineering problems.	A2, Q2, M2, F	C3	2.3H, 2.4L
4	Identify and evaluate the integral of a function.	Q3, M2, F	C6	2.1H, 2.3H
5	Construct the equation of line/plane in space.	A3, F	C5	2.2L, 2.3H

Text Books	Title	Calculus, 8th Edition
	Author	James Stewart
	Publisher	
Reference Books	Title	Calculus, 6th Edition
	Author	Swokowski
	Publisher	
	Title	Calculus, 11th Edition
	Author	Thomas
	Publisher	
	Title	Advanced Engineering Mathematics, 9th Edition
	Author	Erwin Kreyszig
	Publisher	

Week	Course Contents/Topics	Chapter*	CLO*
1	Rules of inequalities and Absolute value, Functions and their graphs	Appendix A	1, 2
2	Basic Concepts of limit, how to evaluate limits, Some examples relating to limits, Continuity of functions, Examples relating to continuity, Types of discontinuities.	2	2
3	Secant line and tangent line, Slope of tangent line and derivative, Horizontal and vertical tangent, Techniques of differentiation	3	3
4	Implicit differentiation and chain rule, Related rates	3	3
5	Role's theorem, Mean Value's Theorem, Geometrical Interpretation of Role's and Mean value theorem	4	3
6	Extreme values of functions and its applications	4	3
7	Horizontal and Vertical Asymptotes	4	3
8	Application of derivatives, Velocity and acceleration of particles, Interval in which body is moving forward and interval in which body is moving backward, Interval of speeding up and slowing down	5	3
9	Rate Of Change And Marginal Analysis, L' Hospital Rule	4	3
10	Fundamental theorem of calculus, Techniques of integration, definite integral	5	4
11	Integration by parts, Reduction Formulae	7	4
12	Improper integrals	7	4
13	Area bounded by the curves	6	4
14	Lines and planes in space, Vector valued functions and space curves	13	5

***Reference book chapters are given in brackets**

Assessment Tools	Weightage
Quizzes (3), Assignments (3)	20.0%
Midterms (I+II)	30.0%
Final Exam	50.0%